

PATTERN COMPLEXITY AND NIVAT’S CONJECTURE

GPT-6 PRO

ABSTRACT. We prove Nivat’s conjecture: a configuration on $\mathbb{Z}^2$ over a finite alphabet with at most mn distinct $m \times n$ patterns, for some positive integers m, n , has a nonzero global period. After labeling the alphabet by rationals, we construct a Laurent polynomial operator that reduces the pattern count by at least the decrease in the window’s area. The construction identifies a line polynomial generating the full annihilator ideal of the difference of two distinct orbit-closure points agreeing on a half-plane. Induction on rectangle area then reduces the conjecture to configurations annihilated by a product of two difference operators. We prove this case using the boundary-extension methods of Cyr–Kra and Szabados. The full proof is formalized in Lean using mathlib.

1. INTRODUCTION

Let A be a finite alphabet and let $c : \mathbb{Z}^2 \rightarrow A$ be a configuration. For a finite set $D \subset \mathbb{Z}^2$, its set of occurring patterns is

$$\text{Pat}_c(D) = \{(c(z + u))_{u \in D} : z \in \mathbb{Z}^2\}, \quad P_c(D) = |\text{Pat}_c(D)|.$$

Write

$$R_{m,n} = \{0, \dots, m-1\} \times \{0, \dots, n-1\}, \quad P_c(m, n) = P_c(R_{m,n}).$$

We call c *periodic* if there is an $h \in \mathbb{Z}^2 \setminus \{0\}$ such that $c(z + h) = c(z)$ for every z . Only one period vector is required.

The Morse–Hedlund theorem says that a bi-infinite word is periodic if and only if, for some n , it has at most n distinct words of length n [MH]. Nivat’s conjecture is the analogous implication for rectangles in two dimensions.

Theorem 1.1. *Let $c : \mathbb{Z}^2 \rightarrow A$ be a configuration over a finite alphabet. If $P_c(m, n) \leq mn$ for some positive integers m, n , then c is periodic.*

Our proof uses two consequences of a low pattern count. After a rational labeling of the alphabet, the occurring patterns satisfy an affine relation; taking a difference gives a polynomial annihilator. This is the starting point of the algebraic approach of Kari and Szabados [KS]. There is also a combinatorial consequence: if adjoining a boundary adds too few patterns, some boundary values are uniquely determined. This observation underlies the generating-set and balanced-set arguments of Cyr and Kra [CK]. Szabados used these methods to prove Nivat’s conjecture for sums of two periodic configurations [Sz].

Kari and Szabados already relate linear dependence of patterns to polynomial annihilators [KS, proofs of Lemmas 6 and 32]. Here we use this relation for differences of patterns with the same polynomial image. The resulting estimate accounts for every coordinate lost when the polynomial is applied to a rectangle, and makes a reduction in area possible. We describe it after fixing notation. Section 5 then gives the two-periodic-summand argument in the form needed to complete the reduction.

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1.1. Notation. An injective labeling of A by rational numbers preserves pattern counts and periods. Except where stated otherwise, configurations below have finite range in $\mathbb{Q}$. This class is closed under rational linear combinations and translations.

For $u \in \mathbb{Z}^2$, let

$$(T^u c)(z) = c(z + u), \quad \Delta_u = T^u - I.$$

These shift and difference operators commute. Set

$$\mathcal{L} = \mathbb{Q}[X^{\pm 1}, Y^{\pm 1}], \quad \mathbf{X}^u = X^{u_1} Y^{u_2}.$$

A Laurent polynomial $f = \sum_u f_u \mathbf{X}^u$ acts by

$$(f(T)c)(z) = \sum_u f_u c(z + u).$$

The sum is finite. We write $\text{supp } f = \{u : f_u \neq 0\}$ and

$$\text{Ann}(c) = \{f \in \mathcal{L} : f(T)c = 0\}$$

for the support of f and the annihilator ideal of c , respectively. Thus $h \neq 0$ is a period of c exactly when $\mathbf{X}^h - 1 \in \text{Ann}(c)$. Our use of forward shifts fixes the sign convention for the polynomial action.

A nonzero vector $v \in \mathbb{Z}^2$ is *primitive* if $\gcd(|v_1|, |v_2|) = 1$. Such a vector extends to a basis v, w of $\mathbb{Z}^2$. Following [KS], a *line polynomial* is a Laurent polynomial whose support contains at least two points and lies on a line. Up to a monomial factor, it has the form $\phi(\mathbf{X}^v)$ for a primitive v and a nonconstant $\phi \in \mathbb{Q}[Z]$ with $\phi(0) \neq 0$.

For later counting arguments, put

$$\delta_c(D) = P_c(D) - |D|.$$

This is the discrepancy of the window D in the terminology of [CK, Section 2.2]. There is one pattern on the empty set, so $P_c(\emptyset) = 1$. Translating a window does not change its complexity. Restriction from a larger window to a smaller one is a surjection between the corresponding sets of occurring patterns.

Let X_c be the closure of $\{T^u c : u \in \mathbb{Z}^2\}$ in the product space $A^{\mathbb{Z}^2}$, where $A = c(\mathbb{Z}^2)$ has the discrete topology. Every finite pattern of a point of X_c occurs in c . Moreover,

$$f \in \text{Ann}(c), \quad x, y \in X_c \implies f(T)x = f(T)y = f(T)(x - y) = 0. \quad (1.1)$$

Both assertions follow by taking limits on finitely many coordinates. The space X_c is compact; sequential compactness can be proved by diagonal extraction over the countable lattice.

1.2. The complexity inequality and the reduction in area. Fix $0 \neq \Phi \in \mathcal{L}$ and a rectangle R , and let

$$S = \{z \in \mathbb{Z}^2 : z + \text{supp } \Phi \subseteq R\}. \quad (1.2)$$

A pattern on R determines the restriction of $\Phi(T)c$ to S . Therefore $P_{\Phi(T)c}(S) \leq P_c(R)$. This inequality alone is insufficient: when $|S| < |R|$, it need not imply $P_{\Phi(T)c}(S) \leq |S|$.

We prove the stronger estimate

$$P_{\Phi(T)c}(S) \leq P_c(R) - |R| + |S| \quad (1.3)$$

under the hypothesis that there are $x, y \in X_c$ with

$$d = x - y \neq 0, \quad \text{Ann}(d) = (\Phi).$$

Here (Φ) denotes the principal ideal generated by Φ . To see why the hypothesis is relevant, let $F : \mathbb{Q}^R \rightarrow \mathbb{Q}^S$ be the map induced by $\Phi(T)$. Every translated restriction of d is the difference of two occurring R -patterns with the same image under F . The exact ideal equality implies that these translated restrictions span *all* of $\ker F$, whose dimension is $|R| - |S|$. A fiber containing k patterns supplies at most $k - 1$ generators for its differences. Summing over the fibers gives (1.3).

The rest of the argument constructs a line polynomial for which this ideal equality holds. A compactness argument gives distinct $x, y \in X_c$ agreeing on a half-plane. A theorem of

Kari–Szabados provides a product of difference operators annihilating $d = x - y$; the form needed here is proved in Appendix A. On configurations vanishing sufficiently far into the half-plane, difference operators transverse to its boundary are injective. Cancelling those factors shows that d has a period qv parallel to the boundary. Section 4 then proves

$$\text{Ann}(d) = (\phi(\mathbf{X}^v)), \quad \phi \mid Z^q - 1,$$

where ϕ is the monic polynomial of least degree satisfying $\phi(T^v)d = 0$. The half-plane condition rules out additional annihilators involving the transverse shift.

Choose a counterexample with a witnessing rectangle of least area, allowing all finite rational alphabets. For $\Phi = \phi(\mathbf{X}^v)$, the set S in (1.2) is a smaller rectangle. Inequality (1.3) makes it a low-complexity window for $c' = \Phi(T)c$. It cannot be empty, since that would give $1 \leq 0$. Minimality therefore makes c' periodic. If $t \neq 0$ is one of its periods, the divisibility $\phi \mid Z^q - 1$ gives

$$\Delta_t \Delta_{qv} c = 0.$$

Section 5 proves that this identity, together with the original rectangular complexity bound, forces c to be periodic. Section 6 assembles these steps into the proof of Theorem 1.1.

2. PATTERN COMPLEXITY UNDER A POLYNOMIAL OPERATOR

We first prove (1.3). For a finite set D , identify $\mathbb{Q}^D$ with the Laurent polynomials supported in D and use the pairing $\langle p, g \rangle_D = \sum_{z \in D} p_z g_z$. The rectangle assumption enters in determining which multiples of Φ can be supported in R .

Lemma 2.1. *Let R be a finite axis-aligned rectangle, let $0 \neq \Phi \in \mathcal{L}$, and set $S = \{z : z + \text{supp } \Phi \subseteq R\}$. Then*

$$(\Phi) \cap \mathbb{Q}^R = \Phi \mathbb{Q}^S, \quad \dim_{\mathbb{Q}}((\Phi) \cap \mathbb{Q}^R) = |S|. \quad (2.1)$$

Proof. Multiplication by Φ sends $\mathbb{Q}^S$ into $\mathbb{Q}^R$. Conversely, suppose that Φg is nonzero and supported in R . The minimum and maximum exponents in each coordinate add under multiplication. For example, regarding the factors as Laurent polynomials in X over the integral domain $\mathbb{Q}[Y^{\pm 1}]$ shows that their leading coefficients have nonzero product. Writing $\max_X f$ for the largest X -exponent in $\text{supp } f$, we obtain

$$\max_X(\Phi g) = \max_X \Phi + \max_X g.$$

The same holds for the minimum exponent and for the Y coordinate.

Write the coordinate intervals of R as $[a_j, b_j] \cap \mathbb{Z}$, and let α_j, β_j be the minimum and maximum j -coordinates of $\text{supp } \Phi$. Every $z \in \text{supp } g$ satisfies

$$a_j - \alpha_j \leq z_j \leq b_j - \beta_j \quad (j = 1, 2).$$

For a rectangle, these inequalities are exactly $z + \text{supp } \Phi \subseteq R$. Hence $g \in \mathbb{Q}^S$. Multiplication by Φ is injective, so its image has dimension $|S|$. The same statements hold when S is empty. $\square$

Theorem 2.2. *Let c have finite rational range, and suppose that $x, y \in X_c$ satisfy*

$$d = x - y \neq 0, \quad \text{Ann}(d) = (\Phi)$$

for some nonzero $\Phi \in \mathcal{L}$. For a nonempty rectangle R , set $S = \{z : z + \text{supp } \Phi \subseteq R\}$. Then

$$P_{\Phi(T)c}(S) - |S| \leq P_c(R) - |R|. \quad (2.2)$$

Proof. Write $\Phi = \sum_u \Phi_u \mathbf{X}^u$. The induced linear map is

$$F : \mathbb{Q}^R \longrightarrow \mathbb{Q}^S, \quad F(p)(z) = \sum_u \Phi_u p(z + u).$$

Its transpose is multiplication by Φ : for $p \in \mathbb{Q}^R$ and $g \in \mathbb{Q}^S$,

$$\langle Fp, g \rangle_S = \sum_{z \in S} \sum_u g_z \Phi_u p(z + u) = \langle p, \Phi g \rangle_R.$$

Consequently $F^\top$ is injective and

$$\dim \ker F = |R| - |S|. \quad (2.3)$$

Let

$$V_R(d) = \text{span}_{\mathbb{Q}}\{T^u d|_R : u \in \mathbb{Z}^2\}.$$

A vector $b = (b_z)_{z \in R}$ is perpendicular to this space exactly when

$$\sum_{z \in R} b_z d(u + z) = 0 \quad \text{for every } u \in \mathbb{Z}^2,$$

that is, when its associated polynomial belongs to $\text{Ann}(d)$. Lemma 2.1 therefore gives

$$V_R(d)^\perp = \text{Ann}(d) \cap \mathbb{Q}^R = (\Phi) \cap \mathbb{Q}^R = \Phi \mathbb{Q}^S = \text{im } F^\top.$$

Taking orthogonal complements yields

$$V_R(d) = \ker F. \quad (2.4)$$

It remains to compare this dimension with the number of occurring patterns. Put $\mathcal{P} = \text{Pat}_c(R)$. Since polynomial operators commute with translations,

$$F(\mathcal{P}) = \text{Pat}_{\Phi(T)c}(S).$$

In each fiber choose a representative, and subtract it from every other pattern in that fiber. The resulting list has

$$|\mathcal{P}| - |F(\mathcal{P})| = P_c(R) - P_{\Phi(T)c}(S)$$

vectors and spans every difference between patterns in the same fiber.

For each u , the patterns $T^u x|_R$ and $T^u y|_R$ occur in c . They have the same image under F , since $\Phi(T)d = 0$. The span of the preceding list thus contains $V_R(d) = \ker F$. By (2.3),

$$|R| - |S| \leq P_c(R) - P_{\Phi(T)c}(S),$$

as required. $\square$

For $\Phi = X - 1$, the map F takes horizontal differences on an $m \times n$ rectangle, and its kernel consists of the patterns constant along each row. Its dimension is n , the number of sites lost on passing to an $(m - 1) \times n$ rectangle. The proof works because the translated restrictions of $x - y$ span all of these row-constant patterns. Knowing only $\Phi(T)(x - y) = 0$ would give an inclusion in the kernel, not the equality needed to count n independent directions.

Corollary 2.3. *In Theorem 2.2, suppose*

$$\Phi = \phi(\mathbf{X}^v), \quad \phi(Z) = a_0 + a_1 Z + \cdots + a_\ell Z^\ell, \quad a_0 a_\ell \neq 0,$$

where $v \neq 0$ and $\ell \geq 1$. If $R = R_{m,n}$ and $P_c(R) \leq mn$, then S is a nonempty rectangle of smaller area and $P_{\Phi(T)c}(S) \leq |S|$.

Proof. The support of Φ contains 0 and ℓv and lies on the segment between them. Since R is a rectangle,

$$S = R \cap (R - \ell v), \quad |S| = \max(0, m - \ell|v_1|) \max(0, n - \ell|v_2|).$$

Theorem 2.2 gives $P_{\Phi(T)c}(S) \leq |S|$. If S were empty, this would say $1 \leq 0$. At least one side length strictly decreases because $v \neq 0$ and $\ell \geq 1$. $\square$

3. AGREEMENT ON A HALF-PLANE

We seek two points of X_c to which Theorem 2.2 can be applied. First we construct distinct points that agree on a half-plane. The product-annihilator theorem then shows that their difference is periodic parallel to the boundary. Neither point need be periodic individually.

3.1. A compactness argument. The existence of half-plane agreement is a symbolic consequence of the results of Boyle and Lind [BL, Theorem 3.7] and Einsiedler, Lind, Miles, and Ward [ELMW, Lemma 2.9]. We give a direct proof that also places a disagreement at the origin.

Lemma 3.1. *If a finite-alphabet configuration c has infinitely many distinct translates, there are $x, y \in X_c$ and a unit vector $\nu \in \mathbb{R}^2$ such that*

$$x(0) \neq y(0), \quad x(z) = y(z) \quad \text{when } \nu \cdot z < 0. \quad (3.1)$$

Proof. For each positive integer j , choose distinct translates x_j, y_j agreeing on the lattice points in the Euclidean ball of radius j . This is possible because there are only finitely many patterns on that ball. Let p_j be a nearest disagreement and put $r_j = \|p_j\| > j$. A nearest disagreement exists because squared lattice norms are nonnegative integers.

By diagonal extraction and compactness of the unit circle, pass to a subsequence such that

$$T^{p_j} x_j \longrightarrow x, \quad T^{p_j} y_j \longrightarrow y, \quad p_j/r_j \longrightarrow \nu, \quad \|\nu\| = 1.$$

Since the alphabet is finite, we may also fix the unequal pair of symbols at the shifted origin. Thus $x(0) \neq y(0)$.

Fix z with $\nu \cdot z < 0$. For all sufficiently large j ,

$$\|p_j + z\|^2 - r_j^2 = 2r_j((p_j/r_j) \cdot z) + \|z\|^2 < 0.$$

By the choice of p_j , the configurations x_j, y_j agree at $p_j + z$. Passing to the limit gives $x(z) = y(z)$. $\square$

Geometrically, the translated agreement balls approach a half-plane with boundary through the fixed disagreement at the origin. The argument gives no agreement on that boundary; nor does it yet imply that the boundary has rational slope.

3.2. Polynomial annihilators. We recall the elementary implication from low complexity to an annihilator [KS, Lemma 5].

Lemma 3.2. *If c is rational-valued and $P_c(D) \leq |D|$ for a nonempty finite set D , then $\text{Ann}(c) \neq \{0\}$.*

Proof. Put $N = |D|$. The at most N augmented vectors $(1, p) \in \mathbb{Q}^{N+1}$, for $p \in \text{Pat}_c(D)$, have a nonzero common orthogonal vector $(\beta, (b_u)_{u \in D})$. Some b_u is nonzero, since otherwise β would also vanish. Hence

$$g = \sum_{u \in D} b_u \mathbf{X}^u \neq 0, \quad g(T)c = -\beta \mathbf{1},$$

where $\mathbf{1}$ is the constant configuration with value one. For any $h \neq 0$, the polynomial $(\mathbf{X}^h - 1)g$ is nonzero and annihilates c . $\square$

We use the following consequence of the Kari–Szabados product-annihilator theorem [KS, Theorem 11 and Corollary 12]. Appendix A gives a proof of this version from their prime-dilation argument. Unlike the full statement of their corollary, it allows repeated directions.

Proposition 3.3 (Kari–Szabados). *If a nonzero finite-range rational configuration c has a nonzero annihilator, then*

$$\Delta_{h_1} \cdots \Delta_{h_s} c = 0 \quad (3.2)$$

for some $s \geq 1$ and nonzero $h_1, \dots, h_s \in \mathbb{Z}^2$.

For finite-range configurations, the removal of repeated factors below is a special case of [KS, Lemmas 9–10]. The following bounded-sequence argument is all that we need.

Lemma 3.4. *Let c be a bounded rational configuration, let $h \neq 0$, and let $s \geq 1$. If $\Delta_h^s c = 0$, then $\Delta_h c = 0$.*

Proof. For each z , set $a_j = e(z + jh)$. If the s th difference of this bounded sequence is zero, its $(s - 1)$ st difference is constant. For $s \geq 2$, a nonzero such constant would make the bounded $(s - 2)$ nd difference grow linearly. The constant is therefore zero. Repetition proves that the first difference vanishes. $\square$

3.3. A period parallel to the boundary. We use the half-plane cancellation argument of Kari and Moutot [KM, Proposition 13 and the proof of Lemma 17]. It removes every difference factor transverse to the boundary; boundedness then deals with any repeated factors in the remaining direction.

Proposition 3.5. *Suppose $d \neq 0$ has finite rational range and a nonzero annihilator. If $d(z) = 0$ whenever $\nu \cdot z < 0$, where $\nu \neq 0$, then there are a primitive $v \in \mathbb{Z}^2$ and $q \geq 1$ such that*

$$\nu \cdot v = 0, \quad T^{qv}d = d.$$

Proof. Let $\mathcal{V}_\nu$ be the vector space of rational configurations e for which, for some $b \in \mathbb{R}$,

$$e(z) = 0 \quad \text{whenever } \nu \cdot z < b.$$

Allowing b to depend on e makes this space invariant under every shift. If $\nu \cdot h \neq 0$, then Δ_h is injective on $\mathcal{V}_\nu$: an h -periodic configuration in this space vanishes, since every h -orbit enters its vanishing half-plane.

Apply Proposition 3.3 to d . The difference operators commute and preserve $\mathcal{V}_\nu$, so each factor transverse to $\nu^\perp$ can be cancelled by this injectivity. Some factor remains, since otherwise $d = 0$. All remaining vectors lie on the line $\nu^\perp$. Choose a primitive lattice vector v along it and write the resulting relation as

$$\prod_{i=1}^r (T^{q_i v} - I)d = 0, \quad q_i \in \mathbb{Z} \setminus \{0\}.$$

Let q be a positive common multiple of the $|q_i|$. In $\mathbb{Q}[Z^{\pm 1}]$, each $Z^{q_i} - 1$ divides $Z^q - 1$. Their product therefore divides $(Z^q - 1)^r$, and $\Delta_{qv}^r d = 0$. Lemma 3.4 gives $\Delta_{qv} d = 0$. $\square$

Cancellation here takes place on $\mathcal{V}_\nu$, not on the space of all configurations, where a difference operator is certainly not injective. The argument is also specifically two-dimensional: all surviving directions belong to a single line. A surviving nonzero lattice vector shows that this line has rational slope.

Corollary 3.6. *If c is a nonperiodic finite-range rational configuration with a nonzero annihilator, then there are $x, y \in X_c$ such that $d = x - y$ is nonzero, vanishes on an open half-plane, and has a period parallel to its boundary.*

Proof. Nonperiodicity makes all translates of c distinct. Apply Lemma 3.1. By (1.1), the difference inherits an annihilator of c , so Proposition 3.5 applies. $\square$

4. THE ANNIHILATOR IDEAL OF THE DIFFERENCE

A period qv gives an annihilator $\mathbf{X}^{qv} - 1$. Theorem 2.2, however, requires a generator of the *whole* annihilator ideal. The vanishing half-plane supplies the additional information: it prevents a recurrence in the transverse direction from introducing a relation not already forced by the minimal polynomial in direction v .

Theorem 4.1. *Let $d : \mathbb{Z}^2 \rightarrow \mathbb{Q}$ be nonzero. Suppose that $T^{qv}d = d$, where v is primitive and $q \geq 1$, and that d vanishes on an open half-plane with boundary parallel to v . Then there is a monic nonconstant $\phi \in \mathbb{Q}[Z]$ such that*

$$\phi \mid Z^q - 1, \quad \text{Ann}(d) = (\phi(\mathbf{X}^v)). \quad (4.1)$$

Proof. Complete v to a lattice basis v, w , orient w toward the other half-plane, and translate if necessary. In these coordinates we can write

$$d(a + q, b) = d(a, b), \quad d(a, b) = 0 \quad (b < 0).$$

Translation leaves the annihilator unchanged. Let Z, W denote the variables for shifts in the v, w directions.

Choose a monic polynomial $\phi(Z)$ of least degree among the nonzero polynomials annihilating d in the first coordinate. Division of $Z^q - 1$ by ϕ leaves an annihilating remainder of smaller degree, which must vanish. Thus $\phi \mid Z^q - 1$. Since $d \neq 0$, this polynomial is nonconstant; since $Z^q - 1$ is square-free over $\mathbb{Q}$, so is ϕ , and $\phi(0) \neq 0$.

We claim that every annihilator $F(Z, W)$ is divisible by ϕ . After multiplication by a Laurent monomial, assume that F is an ordinary polynomial. If some irreducible factor π of ϕ does not divide F , put

$$e = (\phi/\pi)(T^v)d.$$

This configuration is nonzero by minimality of ϕ . It still vanishes on negative rows, and

$$\pi(T^v)e = 0, \quad F(T^v, T^w)e = 0.$$

The purpose of passing from d to e is that every polynomial in Z not divisible by π acts invertibly on the kernel of $\pi(T^v)$, by Bézout's identity.

Write $F(Z, W) = \sum_j f_j(Z)W^j$, let j_* be the largest index with $\pi \nmid f_{j_*}$, and let b_0 be the first nonzero row of e . Evaluate $F(T^v, T^w)e = 0$ on row $b_0 - j_*$. Terms with $j > j_*$ vanish because their coefficients are multiples of π ; terms with $j < j_*$ vanish because they sample rows below b_0 . Thus the nonzero row $u = e(\cdot, b_0)$ satisfies

$$f_{j_*}(T^v)u = 0, \quad \pi(T^v)u = 0.$$

Choose $A, B \in \mathbb{Q}[Z]$ with $A\pi + Bf_{j_*} = 1$. Applying this identity to u gives $u = 0$, a contradiction.

Every irreducible factor of ϕ therefore divides every coefficient of F as a polynomial in W . Squarefreeness gives $\phi \mid F$. Restoring the Laurent monomial proves the same assertion in the Laurent ring. Conversely, every multiple of ϕ annihilates d . The change of lattice basis now yields (4.1). $\square$

The minimal polynomial is necessary, not just a convenient normalization. For $d(i, j) = \mathbf{1}_{\{j \geq 0\}}$, the theorem gives $\text{Ann}(d) = (X - 1)$, although $X^2 - 1$ also annihilates d . On an $m \times n$ rectangle with $m \geq 2$, the translated restrictions of d span an n -dimensional space. The kernel of the map induced by $X^2 - 1$ has dimension $2n$. Choosing the larger period polynomial would therefore destroy the equality (2.4) on which the pattern count depends.

Example 4.2. The boundary term in Theorem 2.2 can be attained. Let

$$d(i, j) = \mathbf{1}_{\{j \geq i\}}, \quad \Phi = XY - 1.$$

The zero configuration belongs to X_d , and Theorem 4.1 gives $\text{Ann}(d) = (XY - 1)$. On $R_{m,n}$, the values of $j - i$ range over $m + n - 1$ consecutive integers. Moving the threshold across these levels gives $m + n$ patterns, including the two constant ones. Since $\Phi(T)d = 0$ and $|S| = (m - 1)(n - 1)$,

$$P_d(m, n) - P_{\Phi(T)d}(S) = m + n - 1 = mn - (m - 1)(n - 1) = |R| - |S|.$$

5. THE CASE OF TWO DIFFERENCE OPERATORS

We prove the result needed when the polynomial image of a minimal-area counterexample is periodic.

Theorem 5.1. *Let $c : \mathbb{Z}^2 \rightarrow \mathbb{Q}$ have finite range. Suppose that*

$$P_c(m, n) \leq mn, \quad \Delta_h \Delta_t c = 0$$

for some $m, n \geq 1$ and nonzero $h, t \in \mathbb{Z}^2$. Then c is periodic. If h, t are nonparallel, a nonzero integer multiple of one of them is a period.

A sum of configurations with periods h, t satisfies this mixed-difference identity. Szabados proved the corresponding two-periodic-summand theorem [Sz, Theorem 1]. The argument below follows his proof: we retain the boundary selection, the ambiguous-extension count, and the finite description of strips, with their sources indicated at the relevant lemmas. The change is the periodicity extension in Lemma 5.8: the mixed-difference identity and a finite recurrence replace the general strip-extension result of [Sz, Lemma 3 and Appendix A.2].

Here is the plan for nonparallel h, t . We find two translates agreeing on a sufficiently wide strip but differing on its next row, unless a multiple of t is already a period. Low complexity supplies a local unique-extension rule on that boundary. The disagreement forces so few patterns inside the strip that Morse–Hedlund gives periodicity there. The local rule then extends periodicity to a few more rows. Only finitely many rows are needed: a difference of c is t -periodic, and their common period consequently extends to the plane.

5.1. One-dimensional lemmas.

Lemma 5.2. *Let $s : \mathbb{Z} \rightarrow A$ have finite range. Suppose that*

$$s_i = s_j \implies s_{i+1} = s_{j+1} \quad (i, j \in \mathbb{Z}).$$

Then s has a positive period at most $|s(\mathbb{Z})|$. The same conclusion holds with $i - 1, j - 1$ in place of $i + 1, j + 1$.

Proof. Successor defines a map on the finite set $s(\mathbb{Z})$. It is surjective, since every occurrence has a predecessor, and hence is a permutation. The sequence follows one cycle of that permutation. Reversing the indices proves the other version. $\square$

For completeness, we give the usual extension-count proof of the form of Morse–Hedlund used below; compare [CK, Section 2.1].

Corollary 5.3 (Morse–Hedlund). *If a finite-range bi-infinite word has at most n words of length n , where $n \geq 1$, then it has a positive period at most n .*

Proof. Let $p(j)$ count its length- j words, with $p(0) = 1$. The counts are nondecreasing. If all n increases up to $p(n)$ were strict, then $p(n) \geq n + 1$. Hence $p(j + 1) = p(j)$ for some $0 \leq j < n$. Restriction of occurring $(j + 1)$ -words to their first j symbols is therefore bijective. Those j symbols determine the following symbol, so the successive $(j + 1)$ -words satisfy the hypothesis of Lemma 5.2. There are $p(j + 1) \leq n$ such words. Their period is a period of the original word. $\square$

The next lemma is the finite-memory version with a periodic parameter. We state the unique-extension property only for inputs that occur; no rule on unused inputs is needed.

Lemma 5.4. *Let $a : \mathbb{Z} \rightarrow A$ have finite range, and let $\eta : \mathbb{Z} \rightarrow E$ be periodic. Suppose that, for some $k \geq 0$,*

$$\eta_i = \eta_j, \quad a_{i+\ell} = a_{j+\ell} \quad (0 \leq \ell < k) \implies a_{i+k} = a_{j+k}.$$

Then a is periodic.

Proof. If $p \geq 1$ is a period of η , the sequence

$$s_i = (i \bmod p, a_i, a_{i+1}, \dots, a_{i+k})$$

has finite range and a uniquely determined successor. Indeed, equality of two such states gives equality of η at the next indices and of the k symbols needed there, and the hypothesis gives the one new symbol. Apply Lemma 5.2. This also covers $k = 0$. $\square$

The output need not inherit a prescribed period of the parameter. For example, a constant parameter can accompany the recurrence $a_{i+1} = 1 - a_i$, whose nonconstant binary solutions have period two. We shall take common multiples of the finitely many row periods that arise, rather than try to preserve a prescribed period at each step.

5.2. Unique extension across part of a boundary. Fix a primitive v and a complementary vector w , so that v, w is a basis of $\mathbb{Z}^2$. The set $\{iv + jw : i \in \mathbb{Z}\}$ will be called row j . Points on a row are ordered by the integer i . These coordinates describe subsets of the original lattice; a rectangle in the original coordinates need not be a rectangle in (i, j) coordinates.

The following is a variant of the edge-removal construction in [Sz, Lemma 2], followed by the unique-extension step from the proof of [Sz, Lemma 4]. We delete a shorter extreme row at each stage. The interpolation argument of [CK, Lemma 2.11] then bounds the lengths of the remaining rows, as needed for Morse–Hedlund.

Lemma 5.5. *Suppose $P_c(R) \leq |R|$ for a nonempty rectangle R . After choosing the sign of w and translating a subset of R , there is a finite set B with bottom row*

$$E = \{0, v, \dots, (e-1)v\}, \quad e \geq 1,$$

and top row $H \geq 0$ such that, with $C = B \setminus E$,

$$P_c(B) \leq |B|, \quad r := P_c(B) - P_c(C) < e. \quad (5.1)$$

If $H > 0$, each integer row from 0 to H contains at least $e-1$ consecutive points of B . Furthermore, for some $0 \leq k < e$, put

$$D_k = C \cup \{0, v, \dots, (k-1)v\}, \quad (5.2)$$

where the second set is empty for $k=0$. Then, for every $u, u' \in \mathbb{Z}^2$,

$$T^u c|_{D_k} = T^{u'} c|_{D_k} \implies c(u + kv) = c(u' + kv). \quad (5.3)$$

Proof. Starting from R , repeatedly delete the nonempty extreme row with fewer sites, choosing either in a tie. Each remaining set is the set of lattice points in the real rectangle intersected with a closed strip. Its nonempty rows are segments of consecutive lattice sites.

Suppose that the current extreme rows have indices $j_- < j_+$ and tangential intervals $[\alpha_-, \beta_-]$ and $[\alpha_+, \beta_+]$. If the shorter row has e sites, both interval lengths are at least $e-1$. For $j_- < j < j_+$, let $\lambda = (j - j_-)/(j_+ - j_-)$. Convexity puts the interval

$$[(1-\lambda)\alpha_- + \lambda\alpha_+, (1-\lambda)\beta_- + \lambda\beta_+]$$

in row j of the real set. Its length is at least $e-1$, so it contains at least $e-1$ consecutive integers. They give points of the current window, proving the row-length bound at each deletion step.

The discrepancy begins at $\delta_c(R) \leq 0$ and ends at $\delta_c(\emptyset) = 1$. At the first step from nonpositive to positive discrepancy, write the deletion as $B \rightarrow C = B \setminus E$. Then

$$0 < \delta_c(C) - \delta_c(B) = e - (P_c(B) - P_c(C)),$$

which proves (5.1). Reverse w if necessary to place the deleted row below the others, and translate its first point to the origin.

For $0 \leq i \leq e$, let $D_i = C \cup \{0, v, \dots, (i-1)v\}$. The e nonnegative integer increments $P_c(D_{i+1}) - P_c(D_i)$ sum to $r < e$. Some increment is zero. At that index k , restriction from occurring D_{k+1} -patterns to occurring D_k -patterns is a bijection. This is (5.3). $\square$

The row-length bound is $e-1$, not e : the endpoints of the interpolated interval need not be integers. In the (v, w) coordinates, the set

$$\{(0, 0), (1, 0), (1, 1), (1, 2), (2, 2)\}$$

illustrates the loss. Its extreme rows have two sites each, but its middle row has only one.

5.3. Two translates agreeing on a strip. Assume in the next lemmas that

$$h = qv, \quad t = av + Mw, \quad q, M \geq 1, \quad a \in \mathbb{Z}, \quad \Delta_h \Delta_t c = 0. \quad (5.4)$$

The identity says that $\Delta_h c$ is t -periodic. Thus the differences between the translates $T^{jt}c$ are h -periodic. As in Szabados's proof [Sz, Section 6], a finite block consequently distinguishes their restrictions to an infinite strip. We express the resulting alternative using Lemma 5.2.

Lemma 5.6. *For every $W \geq M$, either a nonzero integer multiple of t is a period of c , or there are translates x, y of c with*

$$\Delta_h(x - y) = 0, \quad x = y \text{ on rows } 1, \dots, W, \quad x \neq y \text{ somewhere on row } 0. \quad (5.5)$$

Proof. Let S_W be the union of rows $1, \dots, W$, and put

$$\sigma_j = T^{jt}c|_{S_W}, \quad F_W = \{iv + \ell w : 0 \leq i < q, \ 1 \leq \ell \leq W\}.$$

For any j, k , the difference $T^{jt}c - T^{kt}c$ is qv -periodic. Its vanishing on F_W is therefore equivalent to its vanishing on S_W . Restriction to F_W is thus injective on the set $\{\sigma_j : j \in \mathbb{Z}\}$, which is consequently finite. The individual σ_j need not be qv -periodic.

If $\sigma_j = \sigma_k$ always implies $\sigma_{j-1} = \sigma_{k-1}$, Lemma 5.2 gives a positive period p of (σ_j) . Since $W \geq M$, the strips $S_W + jt$ cover $\mathbb{Z}^2$. Equality of the strip restrictions at indices j and $j + p$ then gives $T^{pt}c = c$ everywhere.

Otherwise, choose j, k with $\sigma_j = \sigma_k$ but $\sigma_{j-1} \neq \sigma_{k-1}$. Set $x_0 = T^{jt}c$ and $y_0 = T^{kt}c$. They agree on S_W but differ on $S_W - t$, whose rows range from $1 - M$ to $W - M$. Every positive row in that range already agrees. Let $s \in \{1 - M, \dots, 0\}$ be the largest index of a differing row in this range, and put

$$x = T^{sw}x_0, \quad y = T^{sw}y_0.$$

They differ on row zero. The old rows $s + 1, \dots, 0$ agree by the choice of s , and the old rows $1, \dots, W$ agree by assumption. Since $s \leq 0$, this includes the old rows $s + 1, \dots, s + W$, giving the required agreement on the new rows $1, \dots, W$. Their difference remains h -periodic. $\square$

5.4. Counting extensions and applying Morse–Hedlund. We adapt the argument of Cyr and Kra [CK, Lemma 2.24], also used by Szabados [Sz, Lemma 4]. Grouping finitely many rows into one word converts the extension bound into a word-complexity bound. Here the periodic difference permits a single forward boundary rule. Fix B and the notation of Lemma 5.5, with $H \geq 1$.

Lemma 5.7. *Suppose that translates x, y of c agree on rows $1, \dots, H$, differ on row zero, and have qv -periodic difference for some $q \geq 1$. Then $r \geq 1$, and x restricted to rows $1, \dots, H$ has a common period pv with $1 \leq p \leq r$.*

Proof. The two configurations cannot agree on $E + iv$ for any $i \in \mathbb{Z}$. Such agreement and (5.3) would extend their boundary agreement indefinitely to the right. To obtain the next point, place kv there: the preceding $k < e$ boundary values agree, and all values on the corresponding translate of C agree. For $k = 0$, no preceding boundary values are needed. Since the boundary difference is qv -periodic, vanishing on a right half-line would imply vanishing on the whole boundary, contrary to assumption.

Thus every C -pattern of x encountered under shifts by iv has at least two distinct B -extensions in c . Let $\pi : \text{Pat}_c(B) \rightarrow \text{Pat}_c(C)$ be restriction. The number of patterns with two or more extensions is at most

$$\sum_{p \in \text{Pat}_c(C)} (|\pi^{-1}(p)| - 1) = P_c(B) - P_c(C) = r.$$

At least one such pattern occurs, so $r \geq 1$. Since $r < e$, each row of C has at least $e - 1 \geq r$ consecutive sites.

To obtain one period for all these rows, regard them together as a word over a product alphabet. If $\alpha_j v + jw$ is the first site of row j of C , define

$$b_i = (x((\alpha_j + i)v + jw))_{j=1}^H, \quad i \in \mathbb{Z}.$$

Every length- r word of b is a restriction of one of the translated C -patterns just counted. There are at most r such words. Corollary 5.3 gives a period $p \leq r$ for b , and each coordinate gives the period pv on its entire row. $\square$

The argument uses only a forward unique-extension rule. Agreement propagates to a half-line, and the periodic difference then extends it to the entire boundary. This is the point at which the mixed-difference hypothesis enters the boundary count.

5.5. Extending row periods.

Lemma 5.8. *Assume (5.4) and the unique-extension property (5.3), with $H \geq 1$. If rows $1, \dots, H$ of c are individually periodic in direction v , then a nonzero integer multiple of h is a period of c .*

Proof. Suppose that the H rows immediately above row j are periodic. Set

$$a_i = c(iv + jw), \quad \eta_i = T^{iv+jw}c|_C.$$

A common positive period of those H rows makes η periodic. Property (5.3) says exactly that η_i and $a_i, \dots, a_{i+k-1}$ determine a_{i+k} . Lemma 5.4 shows that row j is periodic as well.

Starting from rows $1, \dots, H$, repeat this argument downward until M consecutive rows are periodic. This requires at most $\max(0, M - H)$ new rows. Choose a positive integer p divisible by q and by a period of each of those M rows, and set $d = \Delta_{pv}c$. The configuration d vanishes on the chosen rows. Writing $p = Lq$, we have

$$d = \sum_{i=0}^{L-1} T^{iqv} \Delta_{qv}c. \quad (5.6)$$

Every summand is t -periodic by (5.4), so d is t -periodic. Translation by t changes the row index by M . Every site can therefore be moved along its t -orbit to one of the M consecutive rows on which d vanishes. Hence $d = 0$ everywhere, and $pv = Lh$ is a nonzero period. $\square$

Only finitely many row periods are combined. Individual periodicity of every row would not by itself imply a common period: for example,

$$c(i, j) = \mathbf{1}_{\{i \equiv 0 \pmod{|j|+2}\}}$$

has no nonzero horizontal period. In the lemma, it is the preexisting t -periodicity of $\Delta_{qv}c$ that propagates a common period from finitely many rows.

Proof of Theorem 5.1. If h, t are parallel, reverse one vector if necessary and take a common positive integer multiple $u = ah = bt$, with $a, b \geq 1$. The telescoping expressions for Δ_u in terms of Δ_h and Δ_t give $\Delta_u^2 c = 0$. Lemma 3.4 yields $\Delta_u c = 0$.

Suppose now that h, t are nonparallel. Write $h = qv$ with v primitive and $q \geq 1$, and complete v to a lattice basis. Apply Lemma 5.5 to the witnessing rectangle to obtain B and a sign of w . Reverse t if necessary so that $t = av + Mw$ with $M > 0$. This preserves the identity because $\Delta_{-t} = -T^{-t}\Delta_t$.

If $H = 0$, then $B = E$ and $P_c(E) \leq e$. Every row word has at most e words of length e . By Corollary 5.3, each row has a period at most e , so $q(e!)v$ is a period of the whole configuration and a multiple of h .

For $H \geq 1$, choose $W \geq \max(H, M)$. Lemma 5.6 either gives a period that is a multiple of t , or gives translates x, y satisfying (5.5). In the latter case, Lemma 5.7 makes rows $1, \dots, H$ of x periodic. The mixed identity and unique-extension property hold for x , so Lemma 5.8 gives it a period that is a multiple of h . A period of a translate is a period of c . $\square$

6. PROOF OF NIVAT'S CONJECTURE

Proof of Theorem 1.1. Suppose a counterexample exists. After rationally labeling its alphabet, choose a nonperiodic finite-range rational configuration c with a witnessing rectangle $R_{m,n}$ of least possible area $N = mn$. The minimum is taken over all finite rational alphabets.

Lemma 3.2 gives a nonzero annihilator of c . By Corollary 3.6, there are $x, y \in X_c$ whose difference d is nonzero, vanishes on a half-plane, and has a period qv parallel to its boundary, with v primitive. Theorem 4.1 gives

$$\text{Ann}(d) = (\Phi), \quad \Phi = \phi(\mathbf{X}^v), \quad \phi \mid Z^q - 1,$$

where ϕ is monic, nonconstant, and has nonzero constant term.

The configuration $c' = \Phi(T)c$ still has finite rational range. Corollary 2.3 gives a nonempty rectangle S of area less than N with $P_{c'}(S) \leq |S|$. Translating S if needed, minimality of N makes c' periodic. Choose a nonzero period t of c' . Then

$$\Delta_t \phi(T^v)c = 0.$$

Multiply by $\psi(T^v)$, where $\psi(Z) = (Z^q - 1)/\phi(Z)$. The result is

$$\Delta_t \Delta_{qv} c = 0.$$

Theorem 5.1, applied to c and its original rectangular complexity bound, makes c periodic, a contradiction. $\square$

The two uses of the alphabet and rectangle in this argument are different. The alphabet may change under $\Phi(T)$, which is why the minimum ranges over all finite rational alphabets. The smaller rectangle remains in the original lattice coordinates, so no complexity assertion on a period sublattice is required.

APPENDIX A. A PROOF OF THE PRODUCT-ANNIHILATOR STATEMENT

We prove Proposition 3.3. The arithmetic step is Lemma 7 of Kari and Szabados [KS], and the resulting product is the one in their Theorem 11. Their Lemma 8 uses the dilation relations in a Vandermonde argument at common zeros. The identity below carries out the corresponding elimination directly in the Laurent-polynomial ring.

Proof of Proposition 3.3. Scale c and a nonzero annihilator separately so that both are integral, and write

$$f = \sum_{i=0}^{k-1} a_i \mathbf{X}^{u_i}$$

with distinct u_i and nonzero integers a_i . Since $c \neq 0$, a monomial cannot annihilate it, so $k \geq 2$. Let

$$K = \max_z |c(z)| \sum_i |a_i|, \quad \rho = K!, \quad f_n = \sum_i a_i \mathbf{X}^{nu_i} \quad (n \geq 1).$$

The positive integer K bounds every value of $f_n(T)c$, independently of n .

If f_n annihilates c and $p > K$ is prime, reduction modulo p gives

$$f_{np} \equiv f_n^p \pmod{p}.$$

Thus every value of $f_{np}(T)c$ is divisible by p and has absolute value at most $K < p$. It vanishes. Starting with $f_1 = f$ and factoring an integer into primes shows that f_n annihilates c whenever $\gcd(n, \rho) = 1$. In particular,

$$f_{1+j\rho} \in \text{Ann}(c) \quad (j \geq 0). \tag{A.1}$$

Put $M_i = \mathbf{X}^{\rho u_i}$. The relations in (A.1) have the form $f_{1+j\rho} = \sum_i a_i \mathbf{X}^{u_i} M_i^j$. We can therefore take linear combinations of them using the coefficients of a polynomial that vanishes at $M_1, \dots, M_{k-1}$. Namely, write

$$H(U) = \prod_{i=1}^{k-1} (U - M_i) = \sum_{j=0}^{k-1} H_j U^j, \quad H_j \in \mathcal{L}.$$

Then

$$\begin{aligned} \sum_{j=0}^{k-1} H_j f_{1+j\rho} &= \sum_{i=0}^{k-1} a_i \mathbf{X}^{u_i} H(M_i) \\ &= a_0 \mathbf{X}^{u_0} \prod_{i=1}^{k-1} (M_0 - M_i). \end{aligned} \tag{A.2}$$

The left side annihilates c . The scalar and monomial factors on the right are units in $\mathcal{L}$. Removing them gives the annihilator

$$\prod_{i=1}^{k-1} (\mathbf{X}^{\rho(u_i - u_0)} - 1).$$

All vectors $\rho(u_i - u_0)$ are nonzero, as required. $\square$

For a trinomial $f = \alpha + \beta X + \gamma Y$, identity (A.2) reads

$$f_{1+2\rho} - (X^\rho + Y^\rho) f_{1+\rho} + X^\rho Y^\rho f_1 = \alpha(1 - X^\rho)(1 - Y^\rho).$$

No separation of the directions of the resulting binomials is needed in Proposition 3.5; boundedness removes repetitions after the transverse factors have been cancelled.

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REFERENCES

- [BL] M. Boyle and D. Lind, *Expansive subdynamics*, Trans. Amer. Math. Soc. **349** (1997), no. 1, 55–102. doi:10.1090/S0002-9947-97-01634-6.
- [CK] V. Cyr and B. Kra, *Nonexpansive $\mathbb{Z}^2$ -subdynamics and Nivat's conjecture*, Trans. Amer. Math. Soc. **367** (2015), no. 9, 6487–6537. doi:10.1090/S0002-9947-2015-06391-0. Numbering refers to arXiv:1208.4090v2 (2013).
- [ELMW] M. Einsiedler, D. Lind, R. Miles, and T. Ward, *Expansive subdynamics for algebraic $\mathbb{Z}^d$ -actions*, Ergodic Theory Dynam. Systems **21** (2001), no. 6, 1695–1729. doi:10.1017/S014338570100181X.
- [KM] J. Kari and E. Moutot, *Decidability and periodicity of low complexity tilings*, Theory Comput. Syst. **67** (2023), 125–148. doi:10.1007/s00224-021-10063-8.
- [KS] J. Kari and M. Szabados, *An algebraic geometric approach to Nivat's conjecture*, Inform. and Comput. **271** (2020), article 104481. doi:10.1016/j.ic.2019.104481. Numbered statements cited here refer to arXiv:1605.05929v1 (2016).
- [MH] M. Morse and G. A. Hedlund, *Symbolic dynamics*, Amer. J. Math. **60** (1938), no. 4, 815–866. doi:10.2307/2371264.
- [Sz] M. Szabados, *Nivat's conjecture holds for sums of two periodic configurations*, in *SOFSEM 2018: Theory and Practice of Computer Science*, Lecture Notes in Comput. Sci., vol. 10706, Springer, Cham, 2018, pp. 539–551. Numbered statements cited here refer to arXiv:1710.05360v1 (2017).